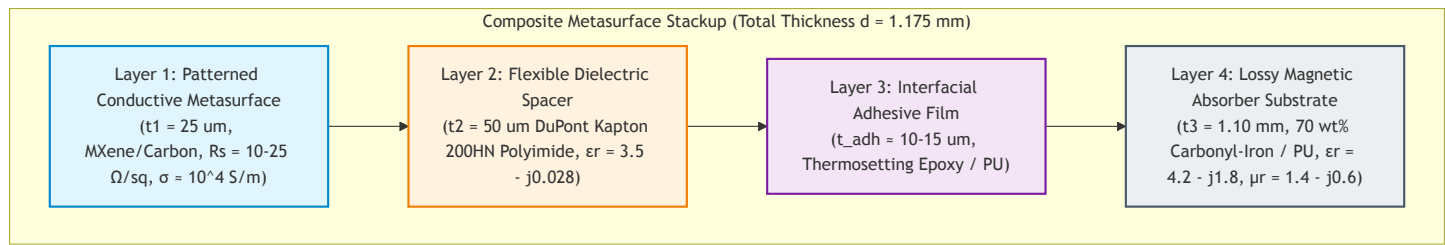


Experimental Fabrication, Assembly & Waveguide Characterization Guide

Project: AI-Driven Equivariant Diffusion Inverse Design for Rozanov-Optimal Metasurface Shielding
Target Frequency Coverage: 8.2 GHz to 18.0 GHz (X-Band & Ku-Band Continuous Sweep)
Target Stackup Thickness: $d = 1.175\text{ mm} \leq 1.20\text{ mm}$
Target Performance: $SE_T \geq 60.0\text{ dB}$, Absorption Ratio $SE_A/SE_T \geq 90.0\%$, Rozanov Metric $\rho_R \geq 0.80$

1. Executive Summary & Physical Stackup Architecture

This standard operating procedure (SOP) details the material preparation, micro-patterning, composite lamination, and Vector Network Analyzer (VNA) waveguide measurement protocols for the inverse-designed absorption-dominant metasurface EMI shields (**Golden Geometry #1** and **Golden Geometry #2**).



2. Bill of Materials (BOM) & Specifications

Component / Layer	Material Description	Recommended Supplier / Grade	Key Target Specification
Conductive Metasurface Pattern	High-conductivity delaminated $\text{Ti}_3\text{C}_2\text{T}_x$ MXene ink OR screen-	In-house MXene synthesis / Dycotec / Henkel Acheson	Concentration $> 25\text{ mg/mL}$, $\sigma > 8,000\text{ S/cm}$, target

Component / Layer	Material Description	Recommended Supplier / Grade	Key Target Specification
	printable carbon/graphene paste		$R_s = 10 - 25 \Omega/\text{sq}$, $\eta = 2.0 \text{ Pa} \cdot \text{s}$
Dielectric Spacer Substrate	DuPont Kapton 200HN Polyimide (PI) film	DuPont Kapton	Nominal thickness $t = 50 \mu\text{m}$ (2 mil), $\epsilon_r = 3.5$, $\tan \delta = 0.008$
Magnetic Filler	Carbonyl-Iron Powder (CIP)	BASF Grade HQ, SM, or EW	Spherical micro-particles, $d_{50} = 2.0 - 4.5 \mu\text{m}$, purity $> 99.5\%$
Polymer Matrix	Flexible Polyurethane (PU) resin OR PDMS (Sylgard 184)	Smooth-On / Dow Corning	Low viscosity prepolymer, thermal curing at 80°C
Interfacial Adhesive	Ultra-thin B-stage thermosetting epoxy/PU film	3M / Toray	Film thickness $10 - 15 \mu\text{m}$, curing at 110°C
Cleaning Solvents	Electronic grade Isopropanol (IPA) & Acetone	Sigma-Aldrich	Purity $> 99.9\%$

3. Unit-Cell Dimensions & Waveguide Array Tiling

The individual unit cell pitch is $P_x \times P_y = 5.715 \text{ mm} \times 5.080 \text{ mm}$. The pattern is tiled into integer arrays that match standard microwave waveguide flanges:

WR-62 Ku-Band Fixture (12.4 - 18.0 GHz)

Inner Aperture: 15.80 mm
x 7.90 mm



Array Tiling: 3 x 2 Unit
Cells
(3 x 5.080 mm) x (2 x
3.810 mm)



Perimeter Conductive
Copper Shim Compensation

WR-90 X-Band Fixture (8.2 - 12.4 GHz)

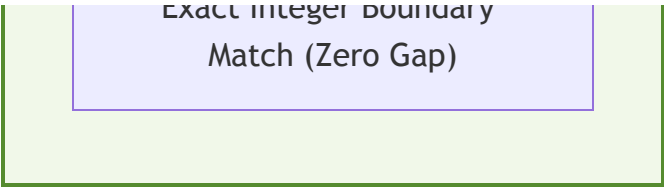
Inner Aperture: 22.86 mm
x 10.16 mm



Array Tiling: 4 x 2 Unit
Cells
(4 x 5.715 mm) x (2 x
5.080 mm)



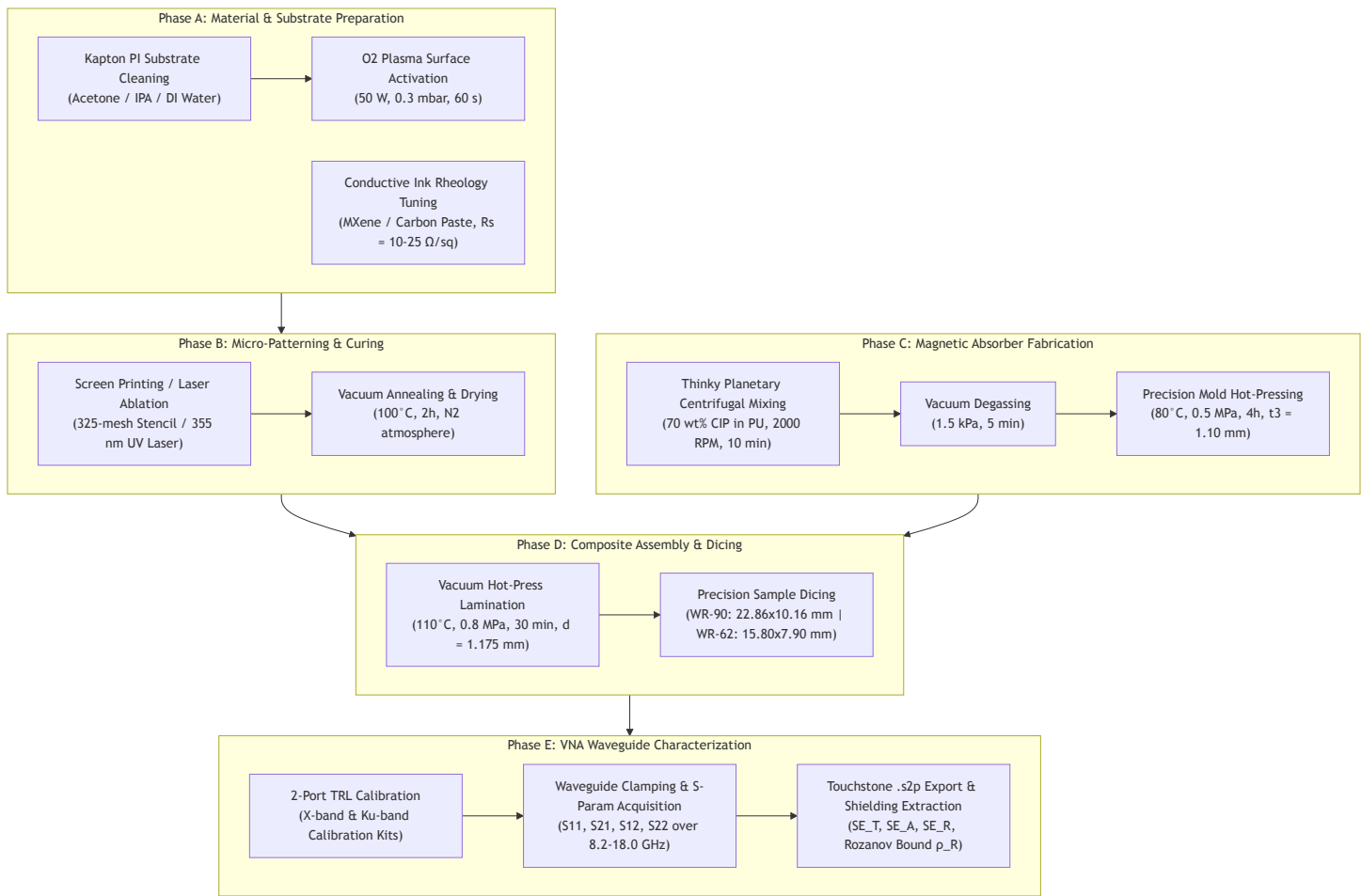
Front Integer Boundary



CAD Vector Files for Mask Generation:

- **Golden Geometry #1 Single Unit Cell:** ../results/exports/golden_geometry_1.dxf
- **Golden Geometry #2 Single Unit Cell:** ../results/exports/golden_geometry_2.dxf

4. Step-by-Step Fabrication Procedure



Detailed Steps:

Step 1: Polyimide Substrate Cleaning & Surface Pre-treatment

1. Cut 50 μm Kapton polyimide sheets into 100 mm × 100 mm working substrates.

2. Ultrasonic cleaning in acetone (5 min), followed by isopropanol (5 min), and deionized water (5 min).
3. Dry under dry nitrogen stream and bake at 120°C for 15 minutes to remove residual moisture.
4. **Surface Activation:** Subject the Kapton sheets to oxygen plasma treatment (50 W, 0.3 mbar O_2 , 60 s) to increase surface energy and promote ink adhesion.

Step 2: Metasurface Conductive Layer Patterning

- **Method A (Screen Printing - Recommended):**
 - i. Prepare a 325-mesh stainless steel screen stencil patterned with the 4×2 array for WR-90 and 3×2 array for WR-62.
 - ii. Align the stencil onto the plasma-treated Kapton sheet with a snap-off distance of 1.5 mm.
 - iii. Print the conductive ink using a 75-durometer polyurethane squeegee at a speed of 40 – 60 mm/s and pressure of 0.2 MPa.
 - iv. Transfer the printed sheet to a vacuum drying oven.
 - v. Dry at 100°C for 2 hours under nitrogen atmosphere (N_2) to prevent MXene oxidation.
- **Method B (UV Laser Direct Ablation - Alternative):**
 - i. Doctor-blade a uniform $25\ \mu\text{m}$ film of conductive ink over the entire Kapton sheet.
 - ii. Vacuum dry at 100°C for 2 hours.
 - iii. Use a 355 nm UV DPSS laser system (e.g., LPKF ProtoLaser) to selectively ablate unwanted conductive regions based on the DXF contour file (laser spot size $\approx 15\ \mu\text{m}$, repetition rate 50 kHz).

Step 3: Carbonyl-Iron Magnetic Absorber Substrate Casting

1. Weigh **70.0 g** of Carbonyl-Iron Powder (CIP) and **30.0 g** of liquid Polyurethane prepolymer (70 wt% CIP loading).
2. Place in a planetary centrifugal vacuum mixer (Thinky Mixer) and mix at **2000 RPM for 10 minutes**.
3. Apply vacuum degassing at **1.5 kPa for 5 minutes** to completely eliminate trapped micro-bubbles.
4. Pour the homogeneous slurry into a precision stainless steel mold with a fixed cavity depth of **1.10 mm**.
5. Close the mold and cure in a heated hydraulic press at 80°C **under 0.5 MPa pressure for 4 hours**.
6. Demold the cured magnetic composite slab ($t_3 = 1.10\ \text{mm} \pm 0.015\ \text{mm}$).

Step 4: Vacuum Hot-Press Stackup Lamination

1. Clean the mating surfaces of the patterned Kapton film and the cured magnetic substrate.
2. Place a 10 μm B-stage epoxy adhesive film between the non-patterned back face of the Kapton sheet and the top face of the magnetic composite.
3. Assemble the stackup between non-stick PTFE release sheets and place inside a heated vacuum laminator.
4. Evacuate the lamination chamber to $< 100 \text{ Pa}$.
5. Ramp temperature to 110°C at $5^\circ\text{C}/\text{min}$ and apply 0.8 MPa **hydraulic pressure**.
6. Hold at 110°C for **30 minutes**, then cool down under pressure to room temperature.
7. Verify final total composite thickness using a digital micrometer: $d_{\text{total}} = 1.175 \text{ mm} \pm 0.020 \text{ mm}$.

Step 5: Precision Sample Dicing

1. Use an automated optical-alignment CNC laser cutter or precision mechanical guillotine to dice the laminated composite into exact waveguide aperture coupons:
 - **WR-90 Sample:** Exactly $22.86 \text{ mm} \times 10.16 \text{ mm}$ (tolerances $+0.00 / - 0.05 \text{ mm}$).
 - **WR-62 Sample:** Exactly $15.80 \text{ mm} \times 7.90 \text{ mm}$ (tolerances $+0.00 / - 0.05 \text{ mm}$).
2. Inspect diced sample edges under an optical microscope to confirm clean vertical cuts with zero delamination.

5. Quality Control & Pre-Test Metrology

Prior to RF testing, execute the following quality inspection steps:

Inspection Item	Equipment / Method	Acceptance Criteria
Sheet Resistance (R_s)	4-Point Probe Station (Keithley / Loresta)	$10 \leq R_s \leq 25 \Omega/\text{sq}$ across 5 sample points
Metasurface Trace Thickness	Optical 3D Profilometer (Bruker / Keyence)	$t_1 = 25 \mu\text{m} \pm 3.0 \mu\text{m}$
Line-Width Resolution	Field-Emission SEM (FE-SEM)	Minimum gap/width $w \geq 150 \mu\text{m}$, edge deviation $\Delta w \leq 15 \mu\text{m}$

Inspection Item	Equipment / Method	Acceptance Criteria
Total Composite Thickness	Precision Digital Micrometer	$d = 1.175 \text{ mm} \pm 0.020 \text{ mm}$
Interfacial Adhesion	Cross-hatch tape test (ASTM D3359)	Grade 4B or 5B (no trace peel-off)

6. Vector Network Analyzer (VNA) Measurement Protocol



Equipment Required:

- 2-Port Vector Network Analyzer (Keysight PNA/ENA or Rohde & Schwarz ZNA) calibrated up to 18.0 GHz.
- Precision WR-90 Waveguide Section & Coaxial Adapters (8.2 – 12.4 GHz, X-band).
- Precision WR-62 Waveguide Section & Coaxial Adapters (12.4 – 18.0 GHz, Ku-band).
- TRL (Thru-Reflect-Line) or SOLT Waveguide Calibration Standards for both bands.
- Flange torque wrench (0.6 N · m).

Measurement Procedure:

1. **Instrument Warm-up:** Power on the VNA and allow 60 minutes for thermal stabilization.
2. **Calibration:**
 - Perform a full 2-port **Thru-Reflect-Line (TRL)** calibration across the X-band (8.2–12.4 GHz, 101 or 201 points) with an IF bandwidth of 1 kHz and output power of 0 dBm.
 - Verify that the calibrated residual directivity is > 45 dB and residual source match is > 40 dB.
3. **Sample Insertion:**
 - Insert the WR-90 metasurface coupon into the waveguide fixture, ensuring the conductive metasurface faces Port 1 (incident port).
 - Tighten waveguide flange screws crosswise with the torque wrench to 0.6 N · m to ensure continuous electrical contact without air gaps.
4. **Data Acquisition:**

- Record full 2-port complex S-parameters: $S_{11}(\omega)$, $S_{21}(\omega)$, $S_{12}(\omega)$, and $S_{22}(\omega)$ in both magnitude (dB) and phase (degrees).
- Repeat measurements for 3 independent test coupons per geometry.

5. Ku-Band Measurement:

- Swap to WR-62 waveguide adapters, re-calibrate via TRL across 12.4–18.0 GHz, insert the WR-62 test coupon, and acquire S-parameters.

6. Data Export:

- Export all datasets as standard Touchstone (.s2p) files:
 - golden_geom1_wr90_sample1.s2p
 - golden_geom1_wr62_sample1.s2p
 - golden_geom2_wr90_sample1.s2p
 - golden_geom2_wr62_sample1.s2p
- Deposit exported .s2p files into the project directory: data/raw/vna_measurements/ .

7. Electromagnetic Shielding Effectiveness Extraction

From the measured complex scattering parameters $S_{11}(\omega)$ and $S_{21}(\omega)$, compute the performance metrics:

1. Reflection Shielding (SE_R):

$$SE_R = 10 \log_{10} \left(\frac{1}{1 - |S_{11}|^2} \right) \quad [\text{dB}]$$

2. Absorption Shielding (SE_A):

$$SE_A = 10 \log_{10} \left(\frac{1 - |S_{11}|^2}{|S_{21}|^2} \right) \quad [\text{dB}]$$

3. Total EMI Shielding Effectiveness (SE_T):

$$SE_T = SE_A + SE_R = 10 \log_{10} \left(\frac{1}{|S_{21}|^2} \right) \quad [\text{dB}]$$

4. Absorption Dominance Ratio:

$$\text{Ratio}_{\text{absorb}} = \frac{SE_A}{SE_T} \times 100\% \quad (\text{Target} \geq 90.0\%)$$

5. Experimental Rozanov Figure of Merit (ρ_R):

$$\rho_R = \frac{\int_{\lambda_{\min}}^{\lambda_{\max}} |\ln |S_{11}(\lambda)|| d\lambda}{2\pi^2 \mu_s d} \quad (\mu_s = 1.4, d = 1.175 \text{ mm, Target } 0.75 \leq \rho_R \leq 0.88)$$

8. Summary Checklist for Experimental Collaborators

- ☐ MXene/carbon ink sheet resistance verified: $10 - 25 \Omega/\text{sq}$ via 4-point probe.
- ☐ Patterned trace thickness verified: $25 \mu\text{m} \pm 3 \mu\text{m}$ on $50 \mu\text{m}$ Kapton polyimide.
- ☐ Magnetic composite substrate thickness verified: $1.10 \text{ mm} \pm 15 \mu\text{m}$.
- ☐ Laminated composite total thickness verified: $1.175 \text{ mm} \pm 20 \mu\text{m}$.
- ☐ Precision dicing verified: $22.86 \text{ mm} \times 10.16 \text{ mm}$ (WR-90) and $15.80 \text{ mm} \times 7.90 \text{ mm}$ (WR-62).
- ☐ 2-Port VNA calibrated via TRL with residual directivity $> 45 \text{ dB}$.
- ☐ Complex S-parameter Touchstone (.s2p) files exported and deposited in
data/raw/vna_measurements/ .